Data Warehousing and ETL- MediLink Health Network

6.1 Introduction

A data warehouse is a centralised repository that integrates data from multiple operational systems and organises it for reporting, analysis, and decision-making. For MediLink Health Network, a data warehouse will consolidate healthcare information from facilities in Kampala, Mukono, Mbarara, and Gulu to provide a unified view of patient activity, service utilisation, and healthcare performance.

The participating facilities may use different operational systems and data formats. Extract, Transform, and Load (ETL) processes will therefore be required to extract source data, standardise and validate it, resolve inconsistencies, and load it into the central analytical environment.

This phase proposes a dimensional data warehouse using a star schema, supported by controlled ETL processes, data-quality checks, and metadata tracking.

6.2 Objectives of the Data Warehouse

The MediLink data warehouse will:

1. Integrate data from the participating healthcare facilities.
2. Provide a consistent structure for cross-facility reporting.
3. Support analysis of patient visits, diagnoses, and healthcare service utilisation.
4. Maintain consistent dimensions for patients, facilities, dates, and services.
5. Preserve historical information for trend analysis.
6. Improve the timeliness and reliability of management reports.
7. Support Ministry of Health reporting requirements, including submission by the fifth day of each month.

6.3 Proposed Data Warehouse Architecture

The data warehouse will follow a layered architecture.

1. Source systems

Kampala · Mukono · Mbarara · Gulu

Patient records, visits, diagnoses and services

2. Extraction and staging

Extract source records and preserve source identifiers

3. Transformation and validation

Standardisation · Data quality · Deduplication · MDM

4. Enterprise data warehouse

Fact tables and conformed dimensions in a star schema

5. Reporting and analytics

Power BI / Tableau · SQL reports · Monthly performance dashboards

Cross-cutting controls: data governance, access control, auditing, metadata and lineage.

This architecture separates operational data collection from analytical processing. It also enables MediLink to validate incoming information before it becomes part of the official reporting dataset.

6.4 Dimensional Modelling and Star Schema

MediLink will use a star schema because it simplifies analytical queries and reporting. A star schema consists of a central fact table connected to descriptive dimension tables.

6.4.1 Fact table

The proposed central fact table is `fact_patient_visit`.

It will store measurable information about patient visits, including:

- Visit count.
- Length of stay, where applicable.
- Service quantity or cost, where reliable financial data is available.
- Links to the relevant patient, facility, date, and service dimensions.

Grain: One row per patient visit or encounter. This grain must be enforced consistently so that a single visit is not counted multiple times during integration.

6.4.2 Dimension tables

Dimension	Example attributes	Purpose
`dim_patient`	Patient key, enterprise patient ID, approved demographic attributes	Enables patient-level analysis with appropriate access restrictions.
`dim_clinic`	clinic key, clinic code, clinic name, district	Supports comparison across healthcare facilities.
`dim_date`	Date key, day, month, quarter, year	Enables daily, monthly, quarterly and annual analysis.

The dimensions will use warehouse surrogate keys to support consistent relationships and historical tracking. Patient identifiers used in analytical systems will be governed carefully to avoid unnecessary exposure of personally identifiable information.

6.5 ETL Process

The ETL process will transfer data from the operational systems into the warehouse through three main stages.

6.5.1 Extract

The extraction process will retrieve the required data from each participating facility, including patient references, visit dates, facility codes, diagnoses, and service records.

Extraction controls will include:

- Recording the source system and extraction timestamp.
- Using incremental extraction where the source supports reliable change tracking.
- Preserving source record identifiers.
- Handling connectivity interruptions through retries and controlled reprocessing.
- Preventing repeated extraction from creating duplicate warehouse records.

6.5.2 Transform

The transformation process will convert source data into the agreed warehouse format.

Transformations will include:

1. Standardising dates and categorical values.
2. Validating required fields and business rules.

3. Normalising phone numbers where they are needed for authorised integration.
4. Mapping facility-specific service and diagnosis codes to approved standards.
5. Resolving patient references through the MDM process.
6. Removing duplicate ingestion events without discarding legitimate repeated visits.
7. Applying dimension-key lookups.
8. Recording rejected records and transformation errors for investigation.

Transformation rules will be version-controlled and documented so that reporting results can be reproduced and audited.

6.5.3 Load

The loading process will populate dimension tables before inserting fact records that reference those dimensions.

The process will:

- Insert or update dimension records.
- Generate or retrieve surrogate keys.
- Load fact records only when required references are valid.
- Use transactions where appropriate to prevent partial loads.
- Record successful, rejected, and failed records.
- Support controlled reprocessing without double-counting visits.

Unknown or unavailable dimension values may be represented using designated warehouse unknown members, provided those records remain traceable and are excluded from analyses where the missing information would make the result misleading.

6.7 Data Warehouse Quality Controls

The following controls will be applied to protect the reliability of the warehouse.

Control	Purpose
Referential integrity checks	Ensure fact records reference valid dimensions.
Duplicate detection	Prevent repeated ingestion from inflating visit counts.
Reconciliation	Compare extracted, loaded, rejected and quarantined record counts.
Business-rule validation	Identify impossible dates, invalid codes and inconsistent measures.
Completeness checks	Identify missing required dimension keys and attributes.
Load monitoring	Detect failed, incomplete or delayed ETL runs.
Audit and lineage records	Trace warehouse records to source systems and transformation steps.

Every ETL run should generate a log containing its run identifier, start and end times, source, extracted row count, loaded row count, rejected row count, and error details.

6.8 Reporting and Business Intelligence

The warehouse will provide a consistent analytical source for Power BI, Tableau, or SQL-based reports.

Proposed reports include:

- Patient visits by facility and reporting period.
- Healthcare service utilisation by category.
- Monthly patient attendance trends.
- Recorded diagnoses by facility and time period.
- Data-quality and rejected-record trends.
- Comparison of facility reporting performance.
- Data reconciliation reports for monthly submissions.

Access to reports will be role-based. Aggregated reporting should be preferred where individual patient identification is unnecessary.

The ETL schedule will be designed to meet the requirement for Ministry of Health reports by the fifth day of each month. The final reporting deadline must account for source submission delays, validation, reconciliation, and approval.

6.9 Implementation Technology

The proposed implementation uses the following technologies.

Component	Technology	Role
Database	MySQL 8.0+	Stores staging data, warehouse dimensions and fact tables.
ETL processing	SQL scripts and scheduled jobs	Extracts, transforms, validates and loads data.
Data visualisation	Power BI or Tableau	Produces dashboards and management reports.
Scheduling	Operating-system scheduler or approved job scheduler	Automates ETL execution.
Monitoring	ETL logs and SQL quality checks	Tracks load success, failures and data-quality exceptions.